"Ethically Unproblematic"? The Unfinished Conversation on Menstrual Blood-Derived Mesenchymal Stem Cells in Neurological Disorders

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The traction received by menstrual-blood derived endometrial mesenchymal stem cells (MenSCs) in the past decade for their accessibility, prevalence, and therapeutic hope has meant they have confronted limited ethical scrutiny compared to alternative sources of stem cells. This has led scientific literature to focus on their application and promising results, ultimately condensing moral conversation into neutrality on the basis of source consent and collection. While progress has been demonstrated in multiple fields of biological sciences and medicine, it is in the treatment of both chronic and acquired neurological conditions that exponential promise has been delivered. Hence, the absence of controversy of embryo destruction, bypassing painful extraction procedures, and considering the pressing race against neurological disorders, has allowed MenSC research to subtly bypass a code of conduct interrogation beyond cell source. This paper brings these unaddressed ethical topics back into the conversation. Discussing how menstrual stigma has presented a structural research barrier, the absence of adequate consent frameworks, an emergence of commercialisation pressures and justice concerns around a gendered therapeutic pipeline; this paper initiates the catch up between ethical discourse and scientific advances.

Introduction

"Nerve paths are something fixed, ended, immobile. Everything may die, nothing may be regenerated." This central dogma founded by Ramón y Cajal (1928) proved to be the initial framework of neuroscience. A rigid claim that has restricted neuroscientific thinking for nearly a century has since been discredited with neuroplasticity, understanding neurogenesis, and most recently stem cell biology. Cells that can recover the ability to differentiate and then integrate have been proven transformative in neurological conditions. Long-standing paradigms on chronic (e.g. Parkinson's disease, Alzheimer's disease, etc.) and acquired (e.g. traumatic brain injury, spinal cord injury, etc.) disorders of the nervous system have been challenged. Conditions such as Alzheimer's and Parkinson's are primarily addressed with treatment plans that rely on slowing progression rather than restoring function, and stem cells have demonstrated therapeutic innovation that could outweigh current medication (Van Bulck et al. 2019; Lindvall and Kokaia 2006). Since their initial discovery, a multitude of stem cell sources have been derived, divided by their specialisation restrictions: adult stem cells (aSC) with multipotency and embryonic stem cells (eSC) that demonstrate pluripotent maturity (Zakrzewski et al., 2019). With controversies of the latter type leading the discussion, the road to implementing stem cells has been paved with ethical obstacles. This has heavily pivoted research focus to aSC.

The multipotent cells have been recovered in various adult tissue sources; notably in endometrial regenerative cells (Gargett et al., 2010). This encompasses at least three functionally distinct progenitor populations: epithelial progenitor cells, endothelial progenitor cells, and mesenchymal stem cells (MSC), which have specifically provoked intrigue with their consistency in regeneration and accessibility. MSC, found in menstrual-blood and the endometrial lining, possess unique multilineage differentiation potential due to the demands of pregnancy and the consistency of menstruation cycles (Baksh et al. 2007). Systematic shedding and growth demonstrate symmetric/asymmetric cell division, making MSC a particularly promising source of stem cell in terms of neuroregenerative therapies (Tajbakhsh et al. 2009). Current studies reveal progress in attenuating neuroinflammation and the ability to differentiate into neural cells (Huang et al. 2014). Menstrual blood-derived mesenchymal stem cells (MenSCs) can be collected from healthy young female individuals, establishing relative accessibility, consent, and abundance; driving the push to expand research in their field. Nonetheless, the exponential therapeutic promise that MenSCs have provided is not met with the necessary discussion on framework or governance. There has been minimal consideration for the persistent stigma around menstrual-blood: consistently concluded that MenSCs are ethically neutral despite several opportunities for ethical deliberation.

Evidently, the relative simplicity of collection: without destruction of the embryo or invasive isolation procedure, and the guaranteed voluntary consent prove MenSCs to be a highly attractive source of stem cells. Occasionally framed as 'waste' has ensured little to no ethical obstruction for the case of MenSCs, rather being praised for repurposing biological material that would otherwise be naturally discarded (Camia and Monti 2025). Nevertheless, considering MenSCs free of ethical dilemma bypasses necessary conversations around broader research ethics. This narrow conception of ethics, restricted to tissue procurement, has proven rushed and incomplete. "'Ethically Unproblematic' The Unfinished Ethical Conversation on MenSCs in Neurological Disorders" completes and extends discussions on sourced based ethics without discrediting therapeutic promise, especially in relation to neurological disease. This paper considers why MenSC research has not prospered proportionally to their innovative results, the so-called 'ick-factor' of menstrual blood, commercialisation, bio-bank consent frameworks and therapeutic applications for neurological conditions. It explores the potential for a distributive justice gap and growing concerns for a gendered therapeutic pipeline, while acknowledging the hope that MenSCs brings to regenerative medicine.

Barriers to Progress: Disparities in MenSC Research

An androcentric approach to science, while recognised and actively addressed, has had repercussions in many aspects of society; with stigma around the menstrual cycle and menstrual blood a persistent by-product. This has led research around menstrual topics to suffer from a stubborn gender disparity, evident when researching the attention and resources allocated (Amaya et al. 2020). Revealing a detrimental impact on the understanding of female biological systems and presenting a clear ethical obstacle that MenSCs investigation has had to overcome. A comparative analysis of published articles between 2008 and 2020 concludes that within the 12 year period, MenSCs accounted for only 0.25% of mesenchymal cell research, with the majority of the first authors being women (Manica et al. 2022). This corroborates the study by Murrar et al. (2021) recognising male-favoured publication bias in research, with high impact research conducted in women being deemed less publishable than research conducted in men (Murrar et al. 2021). Evidently, the lingering effects of gender inequality should be factored in discussions about MenSC research, their progress timeline, and the opportunity to discover novel therapies from these cells.

The residual hesitancy to explore MenSCs, cells that have been consistent in producing compelling results across multiple fields of science, both *in vitro* and *in vivo* models, can be further demonstrated in clinical trial approval. MenSCs have shown the ability to differentiate into cardiomyocytes *in vivo* rat models, into hepatic cells *in vitro*, successfully treat lipopolysaccharide-induced acute lung injury in a murine model, and differentiate most significantly towards glia and neural cells *in vitro*: underscoring their far-reaching promise for regenerative medicine (Hida et al. 2008; Khanjani et al. 2014; Xiang et al. 2017; Azedi et al. 2014, 2017; Kozhukharova et al. 2018; Khanjani et al. 2015; Khanmohammadi et al. 2014; Wolff et al. 2015). Nevertheless, the US government website for tracking clinical trials and their progress (clinicaltrials.gov) showed that only three trials were registered for MenSCs (NCT01496339, NCT01483248, NCT01558908) in 2020 (U.S. National Library of Medicine 2020). Furthermore, as of March 29th, 2026, there were 1,904 studies tagged under mesenchymal stem cells, and only six distinctly focused on those sourced from menstrual blood (NCT05019287, NCT05703308, NCT02095444, NCT01483248, NCT01496339, NCT07131150) (U.S. National Library of Medicine 2026). The display of such a slow incline at attempts to carry out research in MenSCs, while multifactorial, proposes that components beyond scientific feasibility or therapeutic promise are at play. Does chronic social stigma continue to influence the direction scientific progress heads towards?

The Persistence of Stigma as an Ethical and Structural Barrier

Although correlation is famously not causation, the stigma crammed onto the menstrual cycle and its components appears to persist in parallel to the rate of progress, as exposed in a 2023 paper from India. The study clarified how deep cultural opprobrium remained ingrained, hindering the development of biological and biomedical sciences (Kaur et al. 2023). Incorporated to assess the degree of knowledge, perceived attitude and practice of Indian healthcare professionals about MenSCs, the results revealed that although 49% of the respondents knew about MenSCs, only 8.4% had donated menstrual blood before. The lack of donation was attributed to a knowledge gap in the field, as stated by a 47% majority. Across the continents, there is an expectation of cultural variation; however, menstrual taboo has proved to be a unanimous source of consistency (Johnston-Robledo and Chrisler 2020). Despite the intrigue, Kaur et al. (2023) uncover a drastic divergence between awareness and commitment to donation. Therefore, providing an active attempt to increase awareness of MenSCs would benefit the participation of women in donating menstrual blood, encourage researchers to investigate MenSCs therapeutic use, as well as incentivise institutions to allocate funding. This can contribute to dissipating systemic neglect of menstruation-related research whilst continuing to confront societal taboos. The jump between awareness and participation in MenSCs donations illustrates a moral gap that must be addressed simultaneously with respect to the evolution of scientific knowledge.

Institutional Bias in MenSC Research

As a biological secretion, menstrual blood has been overlooked and understudied throughout history (Gunter 2025). Although attempts have been made to correct this disparity, institutions do not always recognise the divide or the importance of addressing it. In 2021, Dr. Federica Helena Marinaro recalled a clear example of the friction that menstrual-related topics provoked, unveiling the extensive criticism she received whilst attempting to publish a paper on the subject to a 9.26 impact factor Springer journal (Manley 2024). The previously published MenSC researcher publicly shared the rejection comments on social media:

> In general, the topic idea is not that novel and could not be accepted as it is, since almost all articles in the literature reported the severe undesirable and toxic effects of menstrual blood and all its constituents (including MenSCs) on the human body. Even in all religions, it is well known that menstrual blood and its MenSCs are extremely very toxic and of very low quality (Marinaro 2021)

Although the journal apologised promptly afterward, the incident is not merely a one-off anecdote. Dr. Marinaro's experience is a suggestive preview at the depth of menstrual stigma has weeded into institutional decisions (Marinaro, 2021). Even funding attributions have presented an additional obstacle, with Imperial College London reproductive immunologist and published researcher Victoria Male claiming that it was easier to obtain funding for immune cells in liver transplantation than for immune cells in the uterus (Khedkar, 2024). Therefore, considering MenSCs to not prompt ethical dialogue discredits the struggle professionals continue to face. The encouraging results, attainability, and abundance of MenSCs have been overshadowed by social stigma, which has attenuated their deserved recognition.

Considering Menstrual Blood as Waste

The term 'waste' has relentlessly been coined by scientific literature to describe menstrual blood. Alternative bodily fluids that could arguably be categorised under the same umbrella; urine, faecal matter, lacrimal fluid, or other secretions have escaped such disregard (Vaughn 2020; Villiger et al. 2018). Practising labelling can consciously or subconsciously influence the framing and emotions evoked when reading. In the case of stem cells derived from female reproductive tissue, such as menstrual blood or umbilical cord, the linguistic bias is not favourable (Das and Sloan 2023). Creating an association with an unnecessary or potentially toxic residue has tainted the lens through which MenSCs are viewed, when in reality they appear more as a treasure trove (Chang and Oatman 2024; Shafiq et al. 2026). The term prevents neither substance from undergoing scientific normalisation, but reinforces the very cultural hierarchies that have contributed to initial scientific neglect.

However, it is not simply donors, larger institutions, or terminology that have imposed shame on the subject of menstruation. Countless professionals and students within the biological, medical or broader scientific fields describe a culturally conditioned so-called 'ick' factor. A Danish medical student named Sara Naseria has publicly commented on her experience with laboratories rejecting menstrual blood, insisting "there [being] an ick factor" (Corbyn 2025). A study carried out by University of St Andrews' Lara Owen reinforces this as a reflection of broader structural patterns (Bildhauer and Owen 2023). Owens demonstrates that menstrual stigma is sustained through socially learnt affective responsiveness, with disgust being the most notable. This positions menstrual blood as a contaminant rather than a neutral bodily fluid, influencing both individual instinctive reactions and institutional willingness to engage. Having to confront the condition of disgust when discussing menstrual materials polarises any ethical neutrality, presenting why ethical deliberations remain relevant.

Collectively, these findings indicate that stigma around menstruation has extended beyond social perception and into regulatory structures of scientific research and education, regardless of therapeutic promise. This has actively shaped the trajectory of what is studied and what is deemed worth studying; raising important ethical questions about the long-lasting impact of ingrained bias on a field of formidable potential. If stigma has pre-emptively determined the topic, terms, and participants of research, then the adequacy of existing consent frameworks deserves equal scrutiny. Not being considered in an ethical conversation would be a disservice to the barriers researchers have repeatedly faced.

Ethical Challenges in MenSCs Bio-banking and Neurological Applications

Recently, there has been a predominant focus on bio-banking: viewed as an opportunity to use autologous MenSCs for the time when age-related medical conditions aggregate and stem cell function and regenerative capacity declines (Vaiciuleviciute et al. 2025). Bio-banking is a term used to describe a scientific facility that stores significant amounts of biospecies, linked to relevant personal and health information, in order to be used in relevant medical research (Annaratone et al. 2021). They are founded on informed consent and an ethical approach to research, in order to provide greater accessibility to scientific and innovation. Despite this, studies have demonstrated that participants lack a proper understanding of consent and the information that this decision entails (Kasperbauer et al. 2022; Eisenhauer et al. 2019). Although personalised medicine does not implicitly violate ethical frameworks, the overwhelming majority of MenSC research is allogeneic; cells donated to unrelated recipients that in return could genetically trace back to the donor. This risk extends to family members, with a lack of understanding of the specific downstream use of MenSCs, leaving donors vulnerable to privacy violations. The threat to confidentiality that stem cell traceability alongside bio-bank longevity provides, imperatively needs to be considered when discussing therapeutic application. Having an intermediate of bio-bank storage between donation and implementation must offer a transparent transition in order to avoid breaches of confidentiality, consent, and an ethically correct outcome.

For 62 years, the Declaration of Helsinki, developed by the World Medical Association (WMA), has served as a code of ethics to guide medical research, involving human participants, and research using identifiable human material or data (World Medical Association 2024). Most recently revised in 2024 to adopt gender neutral language, it additionally recognises participant vulnerability, consent, and bio-banking, among other broader revisions (Parums 2024). The Helsinki Declaration, in its latest rendition, requests transparency about funding and conflicts of interest; it does not govern allogeneic neurological application, genetic traceability of donors, or consent validity in the context of commercial cell line development.

The technicalities regarding consent must be especially examined with neurological conditions as the most clinically significant application. Individuals diagnosed with Alzheimer's, Parkinson's, or spinal cord injuries often demonstrate compromised cognitive capacity, making them less equipped to provide a satisfactory level of informed consent (Alcántar-Garibay et al. 2022; Beccherle et al. 2026; Fang et al. 2020). When a donor gives broad consent for their menstrual sample to be bio-banked, believing themselves adequately informed, donors cannot be confident in anticipating or meaningfully consenting to what their cells will do inside an individual who suffers from such cognitive conditions (Hansson et al. 2006). A vaguer consent term can create a commercially convenient situation where biological donation is not a limiting factor to scientific innovation. This allows for an imbalanced consent pipeline with a donor who provides a broad definition of consent and a recipient who most likely no longer has the neurological capacity to provide consent. Without explicit research guidelines and loose consent requirements, there is a desperate need for ethical intervention.

The question of identity remains a point of contention when considering the neurological application of MenSCs. Beyond procedural gaps and bio-banking parameters, there is a question of whether implanting stem cells would affect personality, self-worth, and if so, to what extent. Personality changes following other transplant procedures have been recorded in recipients for multiple organ sources, including the heart and kidneys (Carter et al. 2024; Bunzel et al. 1992). For cells capable of differentiating into neurons, modulating dopaminergic systems or attenuating neuroinflammation, the assumption that personality and cognitive changes may follow is reasonable and should be recognised in the consent framework. The neuroethics literature has grappled with this possibility of neural grafts and the implications for personal identity (Takala and Buller 2011). Such considerations are not included in the current consent framework that governs the collection of MenSCs, and menstrual blood donors are not informed about the outcome of an intervention of this nature. The possibility of a cognitive or personality change is not inherently cause for ethical alarm; it is the procedural oversight in the conditions of consent for those who donate their cells that poses problems. This inattention is not a minor procedural gap; it is a structural failure to take into account the demands of neurological application. Considering the absence of adequate consent frameworks – which is only emphasised in the context of neurological conditions – and lack of explicit regulation, the opportunity to commercialise MenSCs has arisen.

Commercialisation, Compensation, and the Regulatory Gap

The San Francisco Times uncovers the specific vulnerability of MenSCs to commercialisation in an article centred around the private biotech company Muse. Uncovering the exchange of menstrual blood donation for financial compensation has prompted Lily Chao, 33, to participate conditionally in the research, declaring that "I wouldn't continue donating if not for compensation (Stone 2026)". Although stigma has persisted in deterring both donors and researchers to engage in menstrual blood and MenSCs, it has simultaneously allowed commercial actors to uncover the opportunity for profit within loose legal restrictions. This highlights the pressing speed with which commercial interests can outpace regulatory frameworks. While Muse has appeared in news articles, the nature of the company and Lily Chao's anecdote are not isolated events, but a predictable consequence that arises from an absence of regulations.

Introducing financial compensation to voluntary participation in scientific research is not a novel bioethical debate. The trade between participation and payment has been a longstanding grey zone of research ethics (Largent and Lynch 2017). Principles that the WMA uphold do not directly input their standing in this conversation, requiring: "appropriate compensation and treatment for participants who are harmed as a result of participating in research must be ensured....Free and informed consent is an essential component of respect for individual autonomy... [disclose] incentives for participants (World Medical Association 2024)". These guidelines have established a foundation for participant-based scientific research without addressing the separation between what is deemed an appropriate recognition of consent and donation from undue inducement. This proves particularly problematic for MenSCs as it allows for commercial actors to operate within the ambiguity of the WMA's principles: simply requiring the need to disclose monetary compensation rather than considering it inducement. For women to be persuaded into donating menstrual 'waste' in the name of financial struggle compromise the integrity of their consent. Should donations come from financial necessity?

An aspect of current medical research has transitioned from a moral duty to a moneymaking market, storing a significant untapped profitability (Keshavjee 2004). As life expectancy continuously increases in parallel with modern medicine, this has extended the window of vulnerability to age-related neurodegenerative disorders (Morris 2013). In fact, the Peterson-KFF Health System Track estimates that "In 2024, US life expectancy reached a record high of 79 years, 3.7 years below the comparable country average of 82.7 years (Rakshit et al. 2026)". It is clear that the threat of neurodegenerative diseases increases with age. Meanwhile, the increase in prevalence has not been reciprocated with treatment despite technological advances and billions of dollars worth in funding (Cummings et al. 2018). Decades of research have been shaped around the assumption of neurological rigidity: the brain as static and largely irreparable after injury, a framework Paul Rand critiques in a Harvard-based podcast (Rand and Rust 2025). Nevertheless, accumulating missteps in neuroscientific research, the unanswered demand for treatment, and the growing burden of neurodegenerative disorders have converged into a single gap: the absence of therapies that do more than treat symptoms. It is precisely this vacuum of treatment that has made neurodegenerative disease the most commercially loaded context into which MenSC research is now entering, providing a stream of hope in what appeared to be a drought of progress.

The urgency to prevent being sentenced to discomfort or death simply through the diagnosis of a neurological condition has built up a commercially attractive market, with Alzheimer's disease alone representing a financial burden of over $5 trillion (Liu et al. 2025). Proven to alleviate Alzheimer's disease-like pathology in transgenic mice, or reduce inflammation in Parkinson's disease model rats, MenSCs has been positioned, perhaps prematurely, as the answer to chronic and incurable (Zhao et al. 2018; Li et al. 2023). This insistence on translational momentum has allowed ethical deliberation to be treated as secondary to commercial urgency. Where neurodegenerative diseases were once a life-sentence, stem cell therapies have pushed back on their permanence, and MenSCs have demonstrated an objectively cheap and accessible option. Therapeutic greed has produced the commercially convenient decision to allow an inadequate ethical architecture to persist.

Inevitable Asymmetry: MenSCs and Distributive Justice

The limited data available on organ transplants and donations has still managed to establish a disparity between male and female engagements in the field, with women accounting for up to two thirds of all organ donations (Steinman 2006). This over-representation is seen throughout the world, where women account for more than 50% of living donors, with the exception of Iran (Goldberg, Krause, et al. 2016). Meanwhile, women are more likely to experience delays in access regardless of the conditions of donation (Steinman 2006). These statistics resist biological explanation, with no clinical evidence compelling women to undergo the risks of donation surgery, or explanation to reciprocal effect in organ recipients (Ross and Thistlethwaite 2021). Perhaps, it is better understood as a product of economic dependency, care giving norms, and cultural expectations (Ross and Thistlethwaite 2021). The National Organ and Tissue Transplant Organisation of India reported on the drastic gender disparity that persisted in 2021, with 75% of organ donors being female, and a minority of 20% being organ recipients (Kute et al. 2022). To reiterate, the significant gender disparity between donor and recipient populations is neither absent nor an improving situation; it is a documented structural pattern that merits further ethical scrutiny, especially with the emergence of women-sourced therapeutics. Introducing MenSCs to a system already skewed in favour of male recipients and against female donors demands that distributive justice be treated as a governing constraint on how the therapeutic pipeline is designed, regulated, and administered from the get-go.

Despite a clear focus on source as the predominant ethical premise, little significance has been given to the strictly women supply. This disconnect must first be addressed and then understood in order to factor in the threat of furthering remnants of inequality. MenSCs can only be sourced from healthy women who are at the age of menstruation, making them inherently sex-specific. Consequently, their contribution to a gender-based asymmetry between donor and recipient is inevitable. These eligible donors are drawn from a population that has already been shown to demonstrate higher levels of generosity in healthcare settings, to be subject to care giving expectations, and to have been sold a narrative of repurposing "waste" (Dvorak and Toubman 2013; Anderson and Eifert 1989; Gupta and Pushkala 2020). Therefore, integrating MenSCs as a potential therapeutic tool for devastating neurological conditions into this context would unevenly distribute the benefits in relation to the donations. It may provoke an obligation to donate menstrual blood, discrediting bodily autonomy. To avoid building on the prevalent discrepancy, conversations addressing the ethical framework for MenSCs donations must be confronted. This requires explicit attention to the structural positioning of where and how moral interventions must take place in order to avoid a gendered-therapeutic pipeline at women's expense. Ultimately, even with respect to the source of MenSCs, there is an unfinished discourse to be had regarding the ethical implications of a women-sourced therapeutic.

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Conclusion

Menstrual-blood derived endometrial mesenchymal stem cells (MenSCs) present a unique opportunity for regenerative medicine, with a particularly promising outlook for neurological conditions. Where existing therapies have proved limited, this particular branch of stem cells has uncovered a revolutionary potential for progress. That being said, scientific prospects should not dominate discussions about therapeutic implementation. Especially at the expense of ethical discourse. This paper has exposed how MenSCs are far from ethically neutral; shaped by persistent stigma, insufficient consent frameworks, and the emerging commercial pressures that cannot be dismissed. Vulnerable to existing gender disparities within resource allocation and organ donations, MenSCs merit being addressed beyond source abundance and availability. Appropriate ethical guidelines must be established in order to anticipate decisions that accompany therapeutic implementation and a good practice of research.

The most notorious evidence used to prove ethical neutrality is the source of MenSCs as menstrual blood. A bodily fluid that is associated with 'waste' discarded on a monthly basis. However, it is this source, as one completely dependent on women, that further threatens any gender asymmetry in medicine and healthcare. Without an active effort, the threat of consolidating a sex-based therapeutic pipeline whilst introducing MenSCs-based therapies is imminent. Deliberate intervention must occur to correct the disproportionate foundation at the expense of women, especially when the success of MenSCs places them as a commercially convenient option. The loose consent frameworks and the narrative of recycling 'waste' create misguided and misinformed conditions for donations. Taking a step back from the accelerating scientific progress will allow the ethical dialogue to be filled out to a valuable extent.

Ultimately, recognising that engaging in these conversations does not tarnish the potential that MenSCs represents, is imperative in ensuring they orchestrate a positive impact. They are a brilliant discovery that provides a reliable, abundant, minimally invasive source of multipotent stem cells, with a multitude of applications that scientists are still uncovering. Neurodegenerative intervention of MenSCs alone has changed the way treatment and research has been approached. However, with high rewards comes high risk, and MenSCs provides the perfect example. The history of neuroscience has also served as a guide for the importance of frameworks and curiosity, demonstrating that constant scientific and ethical scrutiny is simply a good practice of science.

It is inevitable that MenSCs have and will continue to tackle structural inequalities and points of moral confrontation. That friction is what propels new branches of scientific thinking and should not discredit the legitimacy of MenSCs-based therapies. However where the issue arises is in the avoidance of these conversations, with ignorance not equating to absence. The concerns this paper addresses are not abstract, but essential to ensuring that progress in this field is not only effective but equitable.

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